

CSC6850 Machine Learning Final Project

Comparative Evaluation of CNN Recurrent Decoders for Optical Chemical Structure Recognition

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01 Problem Motivation



Image-Based Storage

Chemical structures are often stored as images in papers, patents, and databases



Human vs Machine

Images are readable to humans but not directly usable by computers for analysis



OCSR Conversion

Optical Chemical Structure Recognition converts molecule images into SMILES strings



Downstream Uses

SMILES enables search, analysis, database storage, and machine learning applications

02 Why OCSR Is Difficult



Complex Recognition

The model must recognize atoms, bonds, rings, branches, and charges from images



High Sensitivity

A small mistake can completely change the molecule's identity and properties



Sequence Sensitivity

SMILES notation is strictly order-dependent — token position matters



Accuracy Paradox

Correct token accuracy does not guarantee a correct molecule reconstruction

03 Original Plan vs Final Scope

Original Plan

- CNN-based model
- Transformer-based model (SwinOCSR)
- Graph-based model (Atom-level OCSR)
- Multiple datasets: PubChem, EPA, DECIMER

Final Scope

- **CNN encoder + 3 recurrent decoders:**
 - RNN
 - GRU
 - LSTM
- PubChem training
- EPA + DECIMER external testing

04 Related Work



SwinOCSR

Image-to-Sequence OCSR

Swin Transformer encoder with
Transformer decoder for end-to-end
structure recognition



Atom-Level OCSR

Graph Reconstruction

Detects atoms and bonds
individually, then reconstructs the
molecular graph structure



This Project

Simplified Approach

Uses a CNN encoder with recurrent
decoders following the image-to-
sequence paradigm

05 Model Architecture



Models Compared

06 What the Decoder Does

Sequential Token Generation

- Generates SMILES tokens one at a time
- Predicts next token based on:
 - Image features from CNN encoder
 - Previously generated tokens
 - Current hidden state
- Must remember earlier structures (branches, rings) while generating remaining tokens



Example Output



Acetic acid: CC(=O)O

07 RNN vs GRU vs LSTM

RNN

Basic Sequence Memory

Maintains a single hidden state that is updated at each time step. Simple but can struggle with longer sequences due to vanishing gradients.

GRU

Gated Recurrent Unit

Uses update and reset gates to control information flow. The update gate determines how much past information to keep; the reset gate controls past information in new state.

LSTM

Long Short-Term Memory

Uses a memory cell with input, forget, and output gates. More complex architecture provides better handling of long-range dependencies in sequences.

08 Datasets



PubChem

Training & Internal Testing

400 images generated from SMILES via RDKit



EPA CompTox

External Testing

External chemical source for cross-domain evaluation



DECIMER

External Hand-drawn Testing

Hand-drawn molecular images — harder than generated images

PubChem Split

400

Total Images

320

Training Set

80

Test Set

80/20

Split Ratio

seed = 42

09 Evaluation Metrics



Token Accuracy

Measures individual token correctness across the predicted SMILES sequence



Exact Match

Predicted SMILES must exactly match the target string character by character



Canonical Match

Uses RDKit canonicalization to compare chemically equivalent SMILES strings



Valid SMILES Rate

Checks if RDKit can parse the prediction as a valid molecular structure

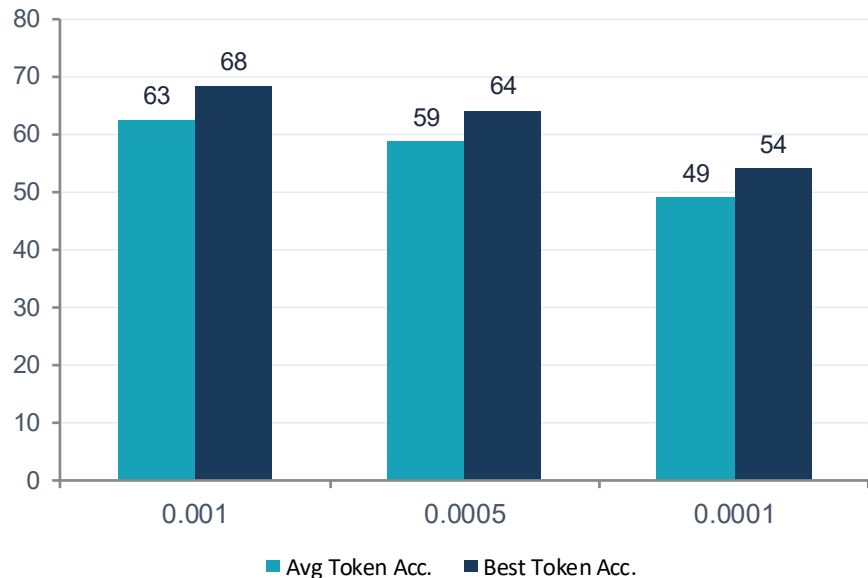
10 Main Results

Dataset	Model	LR	Hidden	Token Acc.	Exact	Canonical	Valid
PubChem	CNN-RNN	0.001	512	64.96%	0.00%	0.00%	0.00%
PubChem	CNN-GRU	0.001	512	68.42%	0.00%	0.00%	100.00%
PubChem	CNN-LSTM	0.001	512	66.79%	0.00%	0.00%	100.00%
EPA	CNN-GRU	0.001	512	41.14%	0.00%	0.00%	100.00%
DECIMER	CNN-GRU	0.001	512	38.62%	0.00%	0.00%	100.00%

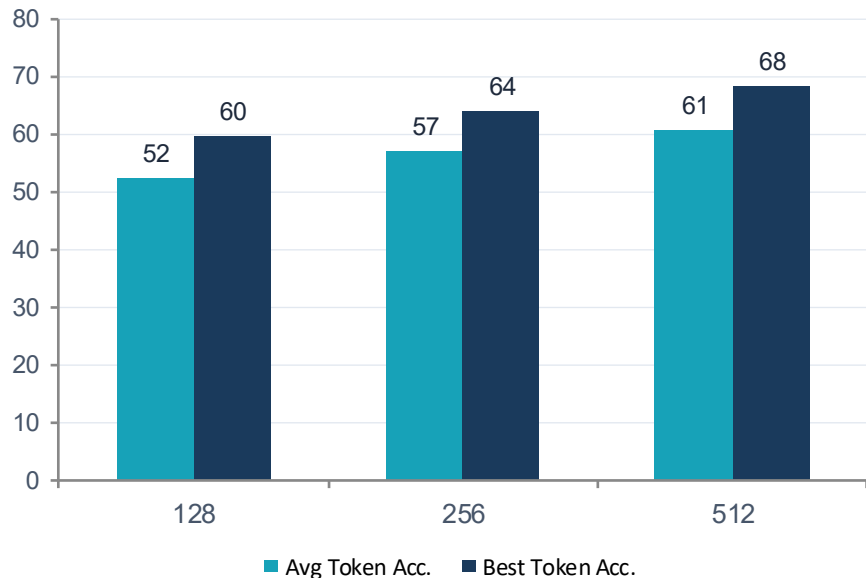
Best result: CNN-GRU at 68.42% token accuracy. Exact and canonical match remained at 0% across all models.

11 Hyperparameter Results

Learning Rate Comparison



Hidden Dimension Comparison



Higher learning rate (0.001) and larger hidden dimension (512) consistently produced the best results.

12 Error Analysis

Common Failure Pattern

```
CCCCCCCCCCCCCCCCCCCCCCC...
```

- This is a valid SMILES string
- But it's usually the wrong molecule
- Model learned common token patterns
- But not global molecular structure



Key Insight

Valid SMILES rate alone is not a sufficient metric. A prediction can be parseable but represent the wrong molecule entirely.

Exact Match	0.00%
Canonical Match	0.00%

13 Hardships & Scope Changes



Original scope was too large for a solo project



Closed-set CNN classifier did not match the OCSR task



Dataset was small (400 images) for sequence generation



External datasets caused significant domain shift



Exact molecule generation is much harder than token prediction

Moving from CNN classifier to image-to-sequence was a key pivot that better matched the OCSR problem.

14 Conclusion



Image-to-Sequence

OCSR should be treated as image-to-sequence translation, not simple classification



CNN-GRU Best

CNN-GRU achieved the highest single result with 68.42% token accuracy



Partial Learning

Token accuracy improved across training, but exact and canonical match remained at 0%



Future Needs

More data, stronger decoding strategies, and better architectures are needed

15 Future Work



Train on a larger PubChem dataset



Add attention between image features and decoder steps



Use beam search instead of greedy decoding



Try SELFIES instead of SMILES



Compare against Transformer-based OCSR



Compare against graph-based atom/bond detection

16 References

1. Xu et al., "SwinOCSR: End-to-end optical chemical structure recognition using a Swin Transformer," 2022.
2. Oldenhof et al., "Atom-level optical chemical structure recognition with limited supervision," CVPR 2024.
3. DECIMER Hand-drawn Molecule Dataset.
4. RDKit: Open-source cheminformatics.
5. PyTorch / torchvision.

Thank You